SQL PROJECT CHALLENGES

Atliq Hardware

CONSUMER GOODS AD_HOC INSIGHTS

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Agenda

- Introduction
- Problem Statement
- Objective
- Market Overview
- Product Overview
- Key Business Questions & Insights
- Strategic Recommendations
- Appendix – SQL Queries

INTRODUCTION

AtliQ Hardwares is a leading computer hardware manufacturer in India with expanding operations in international markets.

With a growing customer base, product portfolio, and distribution network, the company generates large volumes of sales data across regions and channels.

To support faster and smarter decision-making, the management aims to strengthen its data analytics capabilities.

This SQL challenge is designed to evaluate analytical thinking, technical expertise, and the ability to translate data into business insights.

PROBLEM

1. AtliQ Hardwares lacks clear and timely sales insights, making it difficult for management to take fast, data-driven decisions.

2. Limited visibility into product, regional, and customer performance is affecting strategic planning and growth.

OBJECTIVES

- Analyze sales data using SQL
- Identify key trends in product, region, and customer performance
- Generate actionable business insights
- Present findings clearly to support data-driven decision-making

MARKET SEGMENTATION OVERVIEW

Platform

- Brick & Mortar
- E-Commerce

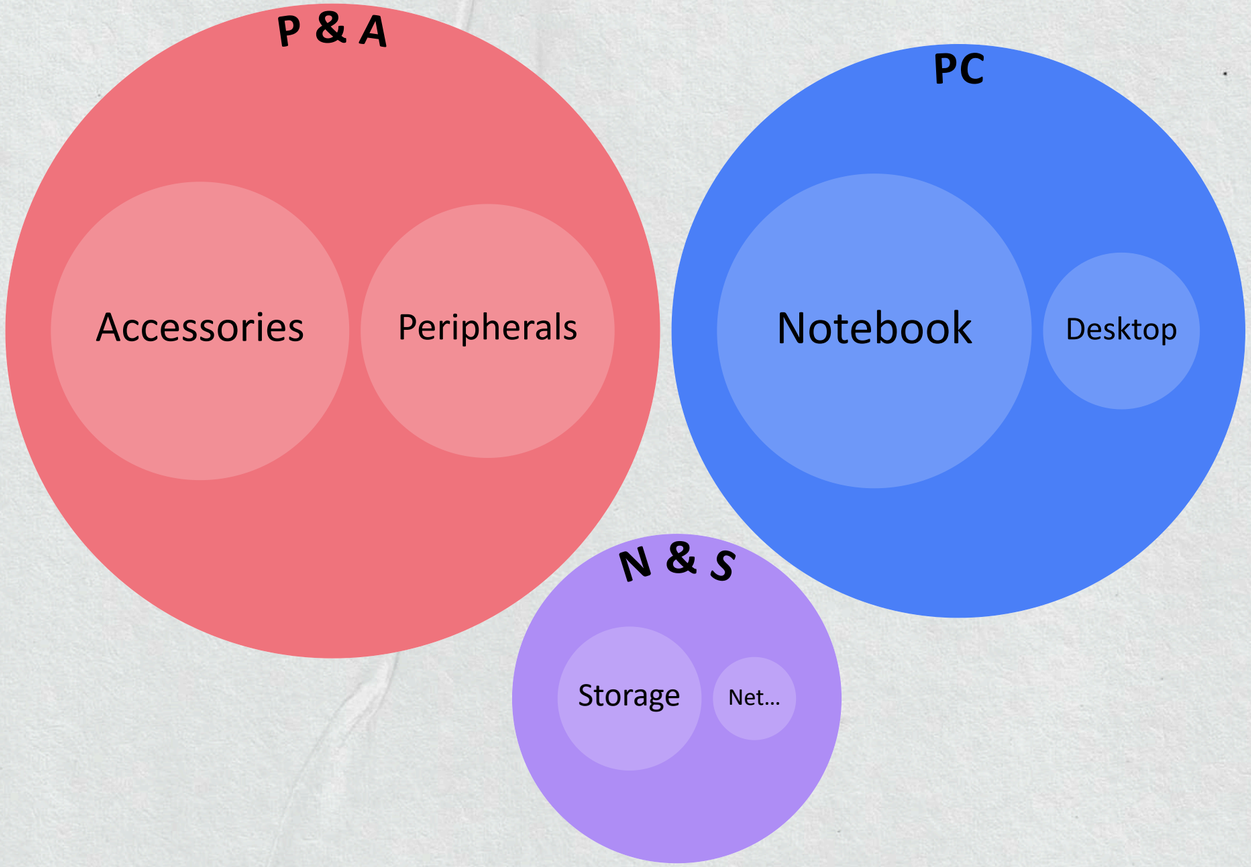
Channel

- Direct
- Distributor
- Retailer

Total 27 country under 4 Region

- APAC
- EU
- NA
- LATAM

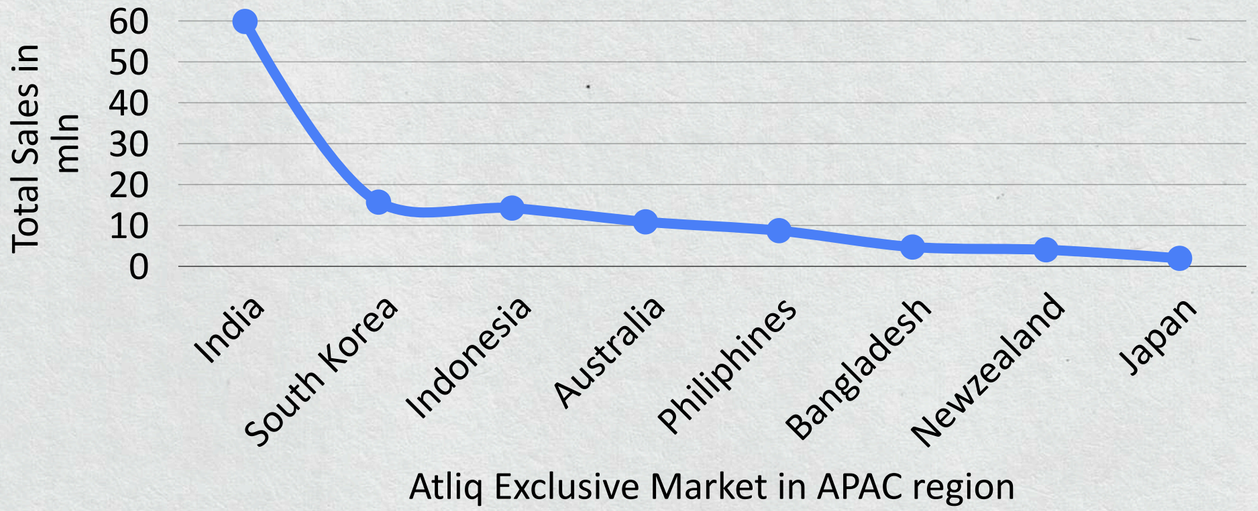
PRODUCT STRUCTURE OVERVIEW



Question 1

PROVIDE THE LIST OF MARKETS IN WHICH CUSTOMER "ATLIQ EXCLUSIVE" OPERATES ITS BUSINESS IN THE APAC REGION.

Australia
Bangladesh
India
Indonesia
Japan
Newzealand
Philiphines
South Korea



Insights

- **India** is the dominant market in the **APAC region**, generating approximately **$59.9M in sales**, which is nearly **four times higher** than the second-largest market, **South Korea.**
- **AtliQ Exclusive's** revenue is heavily concentrated in the Indian market, while all other markets generate less than **$20M**, indicating lower contribution from the rest of the region.

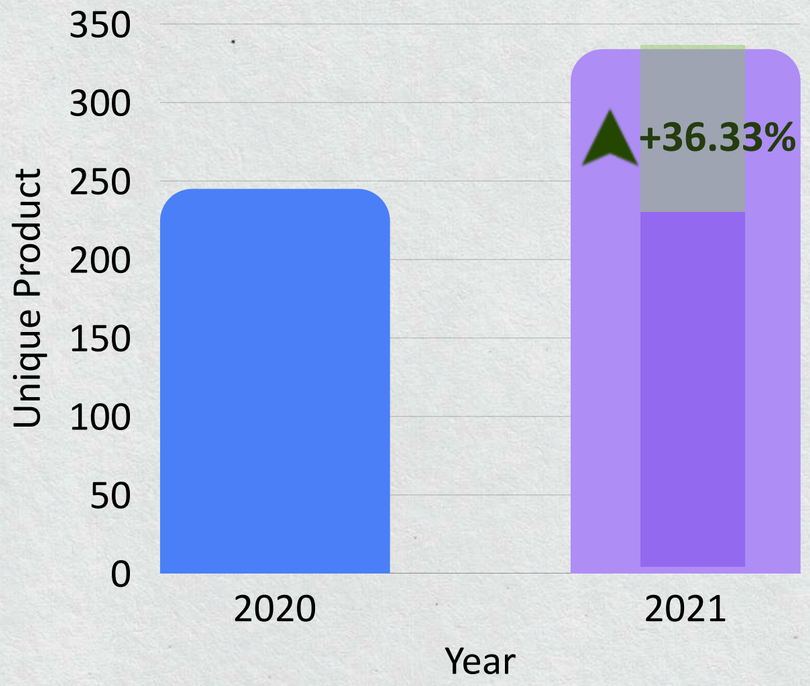
Recommendation

- **T**he company should focus on **increasing sales in South Korea** and **Indonesia** to **reduce dependency** on India and create a more balanced revenue distribution across the region.

Question 2

WHAT IS THE PERCENTAGE OF UNIQUE PRODUCT INCREASE IN 2021 VS. 2020?

Unique Product 2020	Unique Product 2021	Percentage change
245	334	36.33%



Insights

- The company added 89 new unique products in 2021, resulting in a 36.33% year-over-year increase, showing strong product portfolio expansion

.

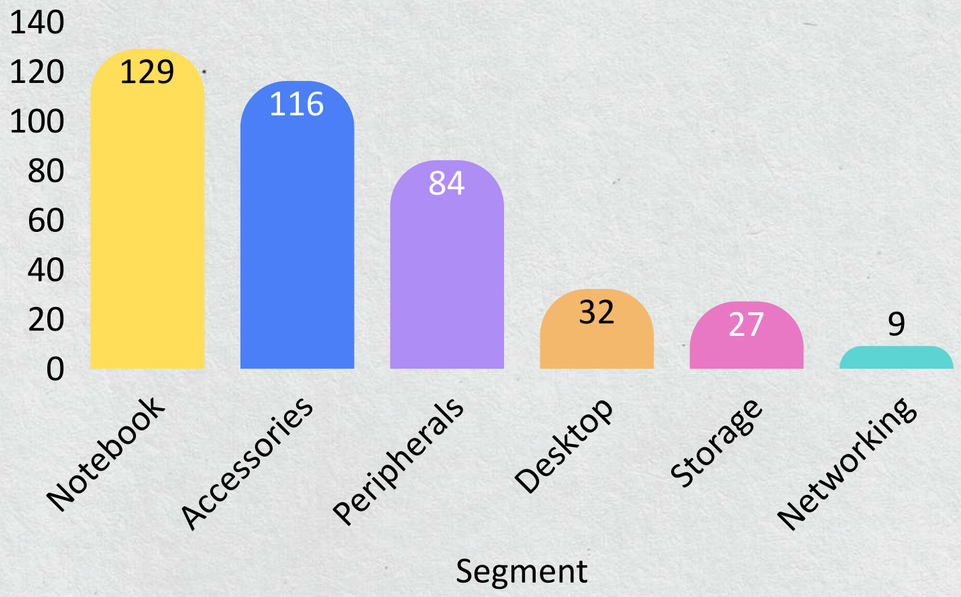
Recommendation

- The company should closely monitor the performance of newly launched products to ensure they contribute positively to revenue and profitability.

Question 3

PROVIDE A REPORT WITH ALL THE UNIQUE PRODUCT COUNTS FOR EACH SEGMENT AND SORT THEM IN DESCENDING ORDER OF PRODUCT COUNTS.

Segment	Product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9



Insights

- Notebook has the highest number of unique products (129), followed by Accessories (116) and Peripherals (84).
- These three segments together contribute about 83% of total products, showing that the portfolio is highly concentrated in these categories.

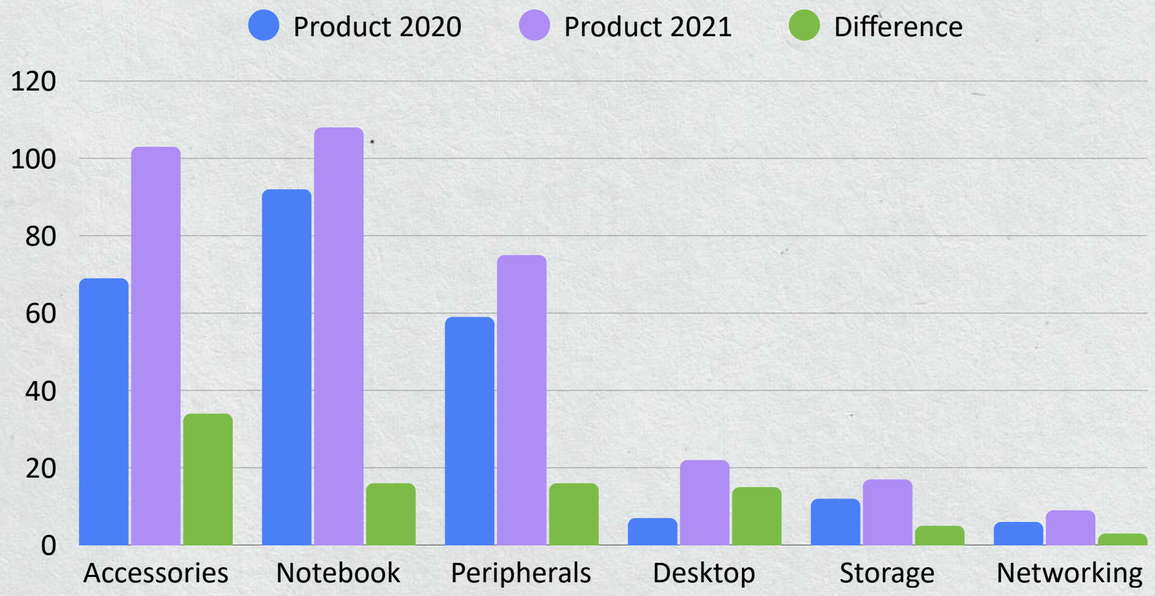
Recommendation

- The company should review the performance of Desktop, Storage, and Networking segments and consider expanding them if they show good revenue and profitability potential.

Question 4

WHICH SEGMENT HAD THE MOST INCREASE IN UNIQUE PRODUCTS IN 2021 VS 2020?

Segment	Product_count_2020	Product_count_2021	Difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3



Insights

- Accessories showed the highest increase in unique products, growing by 34 products from 69 in 2020 to 103 in 2021.
- Notebook and Peripherals also increased by 16 products each, while Networking and Storage saw minimal growth.

Recommendation

- The company should evaluate whether the strong expansion in Accessories is translating into higher revenue and profitability, and ensure that product growth aligns with market demand.

Question 5

GET THE PRODUCTS THAT HAVE THE HIGHEST AND LOWEST MANUFACTURING COSTS.

Product_code	Product	Max/Min Manufacturing Costs
A6120110206	AQ HOME Allin1 Gen 2	240.53
A2118150101	AQ Master wired x1 Ms	0.89

A6120110206	PC	Desktop	Personal Desktop	AQ HOME Allin1 Gen 2	Plus 3

A2118150101	P & A	Accessories	Mouse	AQ Master wired x1 Ms	Standard 1

Insights

- AQ HOME Allin1 Gen 2 (A6120110206) has the highest manufacturing cost at $240.53, indicating it is a premium or high-complexity product.
- AQ Master wired x1 Ms (A2118150101) has the lowest manufacturing cost at $0.89, suggesting it is a low-cost, high-volume product.

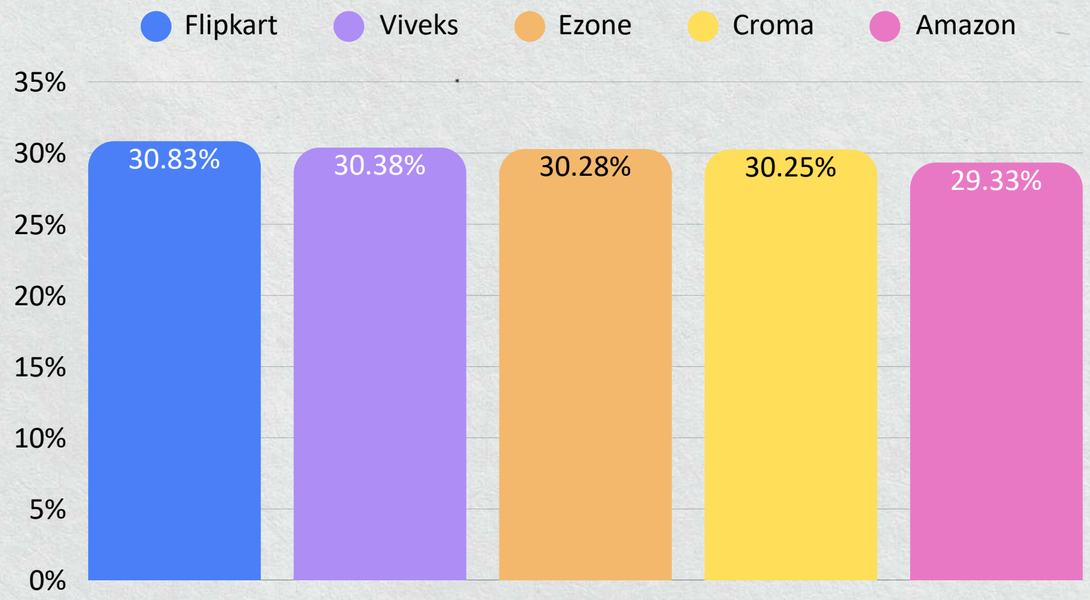
Recommendation

- The company should closely monitor margin performance on high-cost products and ensure low-cost products maintain strong sales volume to maximize overall profitabilit

Question 6

GENERATE A REPORT WHICH CONTAINS THE TOP 5 CUSTOMERS WHO RECEIVED AN AVERAGE HIGH PRE_INVOICE_DISCOUNT_PCT FOR THE FISCAL YEAR 2021 AND IN THE INDIAN MARKET.

Customer	Customer_code	Avg_discount %
Flipkart	90002009	30.83%
Viveks	90002006	30.38%
Ezone	90002003	30.28%
Croma	90002002	30.25%
Amazon	90002016	29.33%



Insights

- Flipkart (90002009) has the highest average discount at 30.83%, followed closely by Viveks and Ezone, indicating aggressive pricing for key customers.
- All major customers receive discounts within a narrow range of 29–31%, showing a largely uniform discount strategy across top accounts.

Recommendation

- The company should evaluate whether these discount levels are affecting overall profitability and consider optimizing discount percentages based on customer performance and margin contribution.

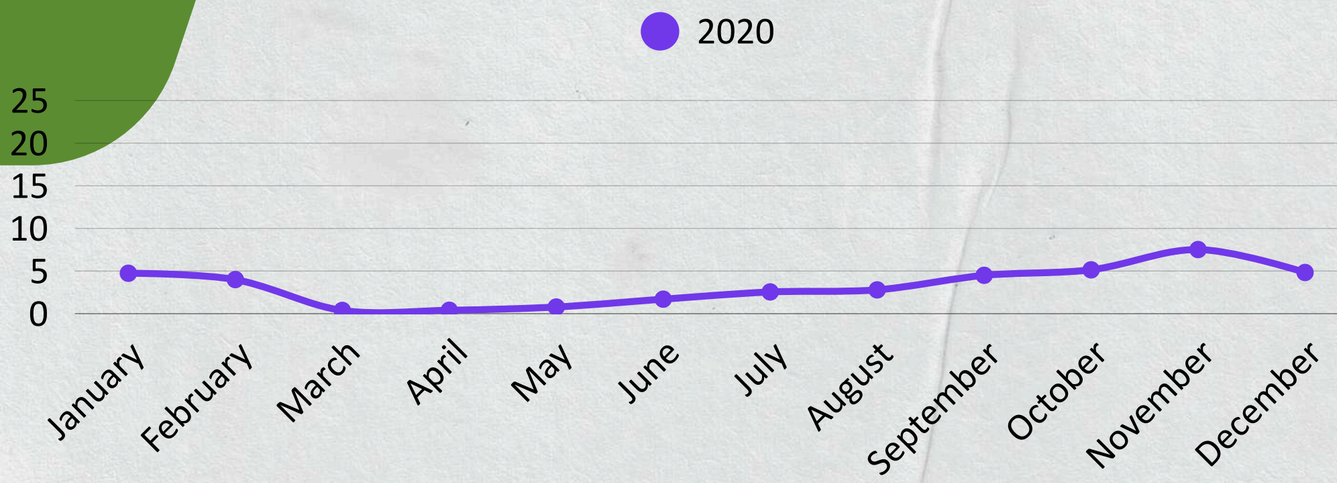
Question 7

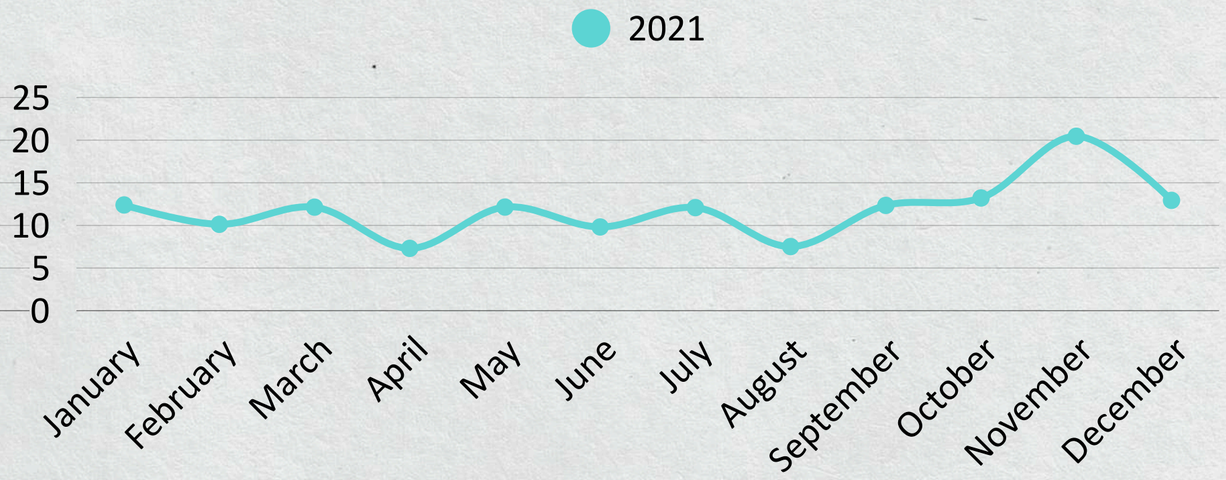
GET THE COMPLETE REPORT OF THE GROSS SALES AMOUNT FOR THE CUSTOMER "ATLIQ EXCLUSIVE" FOR EACH MONTH .

March	2020	38
April	2020	40
May	2020	78
June	2020	170
July	2020	255
August	2020	279
February	2020	400
September	2020	450
January	2020	474
December	2020	483
October	2020	514
November	2020	752
August	2021	718
April	2021	731
June	2021	982
February	2021	1.013
July	2021	1.209
March	2021	1.214
May	2021	1.215
September	2021	1.235
January	2021	1.240
December	2021	1.294
October	2021	1.322
November	2021	2.046

Question 7

GET THE COMPLETE REPORT OF THE GROSS SALES AMOUNT FOR THE CUSTOMER "ATLIQ EXCLUSIVE" FOR EACH MONTH .





Insights

- Sales declined sharply in early 2020, dropping to 38 in March and 40 in April, likely due to the COVID-19 pandemic impact on operations and demand.
- However, sales recovered strongly in the second half of the year, increasing from 170 in June 2020 to 752 in November 2020, showing strong market recovery momentum.
- In 2021, growth accelerated further, with sales reaching 2,046 in November 2021, which is nearly 3x higher than November 2020 (752), indicating significant year-over-year growth.

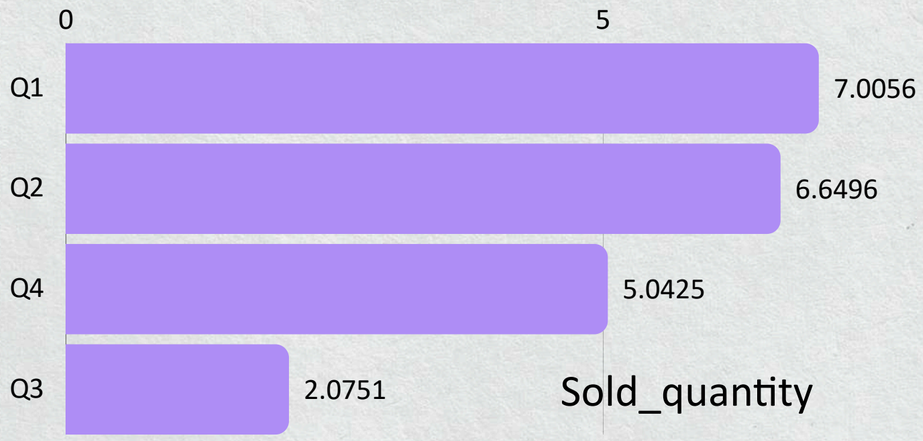
Recommendation

- The company should analyze the factors behind the strong Q4 performance (especially November spikes) and replicate successful sales strategies, promotions, or seasonal planning to sustain growth momentum.

Question 8

IN WHICH QUARTER OF 2020, GOT THE MAXIMUM TOTAL_SOLD_QUANTITY?

Quarter	Sold_quantity
Q1	7.0056
Q2	6.6496
Q4	5.0425
Q3	2.0751



Insights

- Q1 recorded the highest performance at 7.01, followed by Q2 at 6.65, showing strong performance in the first half of the year.
- Q3 dropped significantly to 2.08, making it the lowest-performing quarter, before recovering slightly in Q4 to 5.04.

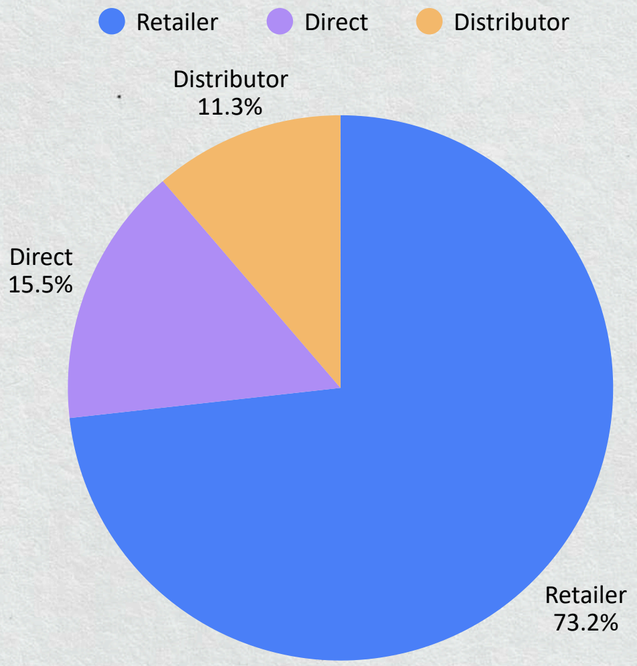
Recommendation

- The company should investigate the sharp decline in Q3 to identify possible seasonal, operational, or demand-related factors and implement strategies to stabilize performance during that period.

Question 9

. WHICH CHANNEL HELPED TO BRING MORE GROSS SALES IN THE FISCAL YEAR 2021

Channel	Gross_sales_mln	% share
Retailer	121.908	73.23
Direct	25.753	15.47
Distributor	18.803	11.30



Insights

- Retailer channel contributes the highest share of gross sales at 73.23%, making it the dominant revenue channel for the company.
- Direct (15.47%) and Distributor (11.30%) channels contribute significantly less, indicating heavy dependence on the Retailer channel.

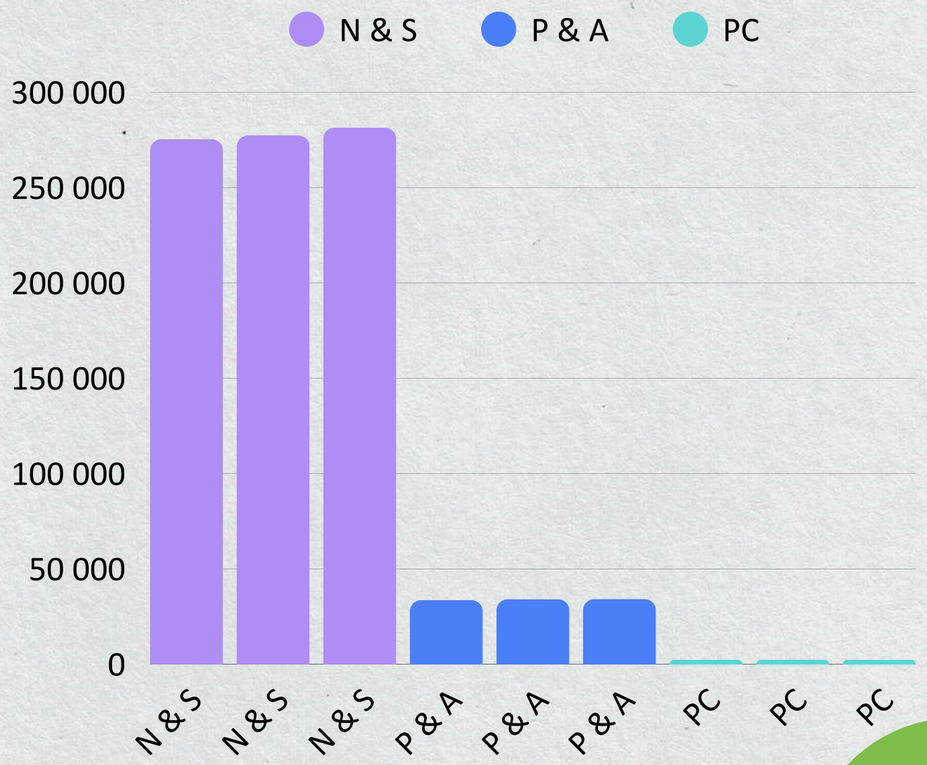
Recommendation

- The company should strengthen relationships with retailer partners while also exploring growth opportunities in Direct and Distributor channels to reduce dependency risk and create a more balanced revenue mix.

Question 10

GET THE TOP 3 PRODUCTS IN EACH DIVISION THAT HAVE A HIGH TOTAL_SOLD_QUANTITY IN THE FISCAL_YEAR 2021?

division	product_code	product	total_sold_quantity	Top 3
N & S	A7219160201	AQ Wi Power Dx2	275328	1
N & S	A7220160203	AQ Wi Power Dx2	277299	2
N & S	A7321160301	AQ Wi Power Dx3	281363	3
P & A	A3920150304	AQ LION x3	33523	1
P & A	A3718150102	AQ LION x1	34022	2
P & A	A3718150105	AQ LION x1	34080	3
PC	A6018110106	AQ Home Allin1	2281	1
PC	A6119110202	AQ HOME Allin1	2285	2
PC	A6119110204	AQ HOME Allin1	2286	3



Insights

- Sales are heavily dominated by the N & S division, with top products selling above 275K units, which is significantly higher than P & A (around 34K units) and PC (around 2K units).
- The PC division shows very low sales volume compared to other divisions, indicating limited market demand or lower product penetration.

Recommendation

- The company should analyze profitability across divisions to ensure high-volume N & S products are also margin-efficient, while developing targeted strategies to improve sales performance in the PC division if it has growth potential.

STRATEGIC RECOMMENDATIONS

- Reduce revenue concentration risk by expanding growth in underperforming markets and channels.
- Strengthen high-performing divisions (N & S) while evaluating profitability of low-performing divisions (PC).
- Monitor new product performance to ensure portfolio expansion drives profitability.
- Leverage strong seasonal quarters (Q4) by replicating successful sales and marketing strategies.
- Optimize discount strategy to balance competitive pricing and margin protection.

SQL QUERY

1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

```
SELECT DISTINCT
   market,
   SUM(sold_quantity * gross_price) / 1000000 AS total_sales
FROM
   dim_customer
      JOIN
   fact_sales_monthly f USING (customer_code)
      JOIN
   fact_gross_price gp ON f.product_code = gp.product_code
      AND f.fiscal_year = gp.fiscal_year
WHERE
   customer LIKE '%atliq exclusive'
      AND region LIKE '%APAC%'
GROUP BY 1
ORDER BY total_sales DESC;
```

SQL QUERY

2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields, unique_products_2020 unique_products_2021 percentage_chg

```
with unique_count as ( SELECT
   COUNT(DISTINCT CASE
         WHEN fiscal_year = 2020 THEN product
      END) AS unique_2020,
   COUNT(DISTINCT CASE
         WHEN fiscal_year = 2021 THEN product
      END) AS unique_2021
FROM
   fact_sales_monthly
      JOIN
   dim_product USING (product_code)
WHERE
   fiscal_year IN (2020 , 2021)
)
SELECT
   *,
   CONCAT(ROUND((unique_2021 - unique_2020) * 100 / unique_2020,
               2),
         '%') AS pct_change
FROM
   unique_count
```

SQL QUERY

3. Provide a report with all the unique product counts for each segment and sort them in descending order of roduct counts. The final output contains 2 fields, segment product_count

```
SELECT
    segment, COUNT(DISTINCT product_code) AS product_count
FROM
    dim_product
GROUP BY segment
ORDER BY product_count DESC
```

SQL QUERY

4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields, segment product_count_2020 product_count_2021 difference

```
with  product_count_2020 as (SELECT
   segment, COUNT(DISTINCT product_code) AS product_count2020
FROM
   dim_product
      JOIN
   fact_sales_monthly USING (product_code)
WHERE
   fiscal_year = 2020
GROUP BY segment
ORDER BY product_count2020 DESC
),

product_count_2021 as (SELECT
 segment, COUNT(DISTINCT product_code) AS product_count2021
FROM
 dim_product
 JOIN
 fact_sales_monthly USING (product_code)
WHERE
 fiscal_year = 2021
GROUP BY segment
ORDER BY product_count2021 DESC
)

SELECT
   *, (product_count2021 - product_count2020) AS difference
FROM
   product_count_2020
      JOIN
   product_count_2021 USING (segment)
ORDER BY difference DESC
```

SQL QUERY

5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields, product_code product manufacturing_cost

```
with maxium_man_cost as (SELECT
    product_code, product, manufacturing_cost
FROM
    fact_manufacturing_cost
        JOIN
    dim_product USING (product_code)
ORDER BY manufacturing_cost DESC
LIMIT 1),
 minimum_man_cost as (SELECT
    product_code, product, manufacturing_cost
FROM
    fact_manufacturing_cost
        JOIN
    dim_product USING (product_code)
ORDER BY manufacturing_cost ASC
LIMIT 1)
 select * from maxium_man_cost union
 select * from minimum_man_cost
```

SQL QUERY

6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields, customer_code customer average_discount_percentage

```
SELECT
   c.customer,
   f.customer_code,
   CONCAT(ROUND(AVG(f.pre_invoice_discount_pct) *
100, 2),
         '%') AS average_discount_pct
FROM
   dim_customer c
      JOIN
   fact_pre_invoice_deductions f USING (customer_code)
WHERE
   market = 'India' AND fiscal_year = 2021
GROUP BY 1 , 2
ORDER BY AVG(f.pre_invoice_discount_pct) DESC
LIMIT 5
```

SQL QUERY

7. Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions.

```
SELECT
   MONTHNAME(date),
   f.fiscal_year,
   ROUND(SUM(sold_quantity * gross_price) / 1000000,
         2) AS Gross_sales_amt_mln
FROM
   fact_sales_monthly f
      JOIN
   fact_gross_price gp ON f.product_code =
gp.product_code
      AND f.fiscal_year = gp.fiscal_year
      JOIN
   dim_customer c ON f.customer_code =
c.customer_code
WHERE
   c.customer = 'Atliq Exclusive'
GROUP BY 1 , 2
ORDER BY fiscal_year , Gross_sales_amt_mln
```

SQL QUERY

8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity, Quarter total_sold_quantity

```
SELECT
   CASE
      WHEN MONTH(date) IN (9 , 10, 11) THEN 'Q1'
      WHEN MONTH(date) IN (12 , 1, 2) THEN 'Q2'
      WHEN MONTH(date) IN (3 , 4, 5) THEN 'Q3'
      ELSE 'Q4'
   END AS Quarter,
   SUM(sold_quantity) / 1000000 AS total_qty
FROM
   fact_sales_monthly
WHERE
   fiscal_year = 2020
GROUP BY Quarter
ORDER BY total_qty DESC
```

SQL QUERY

9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel gross_sales_mln percentage

```
with gross_sales_amt as (
SELECT
   c.channel,
   ROUND(SUM(sold_quantity * gross_price / 1000000),
         2) AS gross_sales_mln
FROM
   dim_customer c
      JOIN
   fact_sales_monthly f ON c.customer_code =
f.customer_code
      JOIN
   fact_gross_price gp ON f.product_code =
gp.product_code
      AND f.fiscal_year = gp.fiscal_year
WHERE
   f.fiscal_year = 2021
GROUP BY 1)
select channel, gross_sales_mln,
round((gross_sales_mln*100/ sum(gross_sales_mln)
over() ),2)  as pct  from gross_sales_amt
group by channel
order by pct desc
```

SQL QUERY

10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields, division product_code product total_sold_quantity rank_order

```
WITH prod_ranking AS (SELECT
division,
 f.product_code,
 product,
 sum(sold_quantity) as total_qty,
dense_rank() over(partition by division order by
sum(sold_quantity)) as ranking
FROM fact_sales_monthly f
JOIN
 dim_product p on f.product_code = p.product_code
WHERE fiscal_year = 2021
GROUP  BY 1,2,3)
SELECT  * from prod_ranking
```

THANK YOU